PhD Comprehensive Examination
on
Development and Simulation of AI-Integrated UAV Platforms for Advanced Landmine Detection with GPR and Metal Detectors

(Tentative in)

National Institute of Technology Raipur
Department of Information Technology

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Doctoral Guidance Committee (DGC)

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Contents

Course Work Requirement: Completed

R.12.0 COURSEWORK/CREDIT REQUIREMENTS

R.12.1. All students of Ph.D. program are required to earn course-credits as prescribed by the supervisor(s) and approved by the DGC/DRC. All students shall complete credit requirements through course work as specified in the table below.

S. no.	Candidates having qualifications	Credit requirement
1	M. Tech. or Equivalent; M Arch/ M Plan or Masters in any relevant discipline of Architecture OR MBA with B. Tech. (for management only);	12 credits
2a	M.Sc./MCA/MA or equivalent	16 credits
2b	M.Sc./MCA/MA with M. Phil.	12 credits

Total credits to be completed: 12

S. No.	Subject	Semester	Types of Course	Credits Earned
1.	Research Methodology and Technical & Scientific Application	First	Regular	4
2.	Internet of Things	First	Seminar	4
3.	Seminar	Second	Minor Project	4

Introduction to Landmine Detection

Landmines and UXOs endanger lives in post-conflict areas. Traditional detection is slow and risky, while UAVs offer safer, faster surveying. Equipped with Ground Penetrating Radar and metal detectors, drones can detect both metallic and non-metallic mines efficiently

Fig. 1 : Drone integrated metal detection system

Introduction to Landmine Detection

- UAV-mounted sensors face challenges like altitude-related antenna coupling, vibrations, motion noise, and effects of soil, moisture, and vegetation.
- Researchers optimize sensor setup, flight height, antenna stability, and signal processing to improve accuracy.
- Artificial Intelligence enhances detection by automating recognition, reducing false alarms, and improving classification.
- Deep learning models such as CNNs and RNNs effectively analyze radar and metal detector data for buried mine patterns.
- Simulation and modeling allow safe testing of sensors under varied soil and environmental conditions.

History of Detecting Landmines

- → **Pre-2000s:** Manual probing, dogs, and metal detectors—slow and risky.
- → **2000s:** Early remote sensing for minefield mapping.
- → **2010–2015:** UAVs introduced for aerial surveillance.
- → **2016–2019:** Integration of GPR and metal detectors on drones.
- → **2020–2022:** AI and deep learning improve detection accuracy.
- → **2023–Present:** Autonomous, AI-integrated UAV enable efficient humanitarian demining

Detection Workflow

Abnormal Sensor Readings: Sudden change in Ground Penetrating Radar (GPR) reflections or signal intensity.

Metal Detector Alerts: Spike in electromagnetic response indicating metallic presence.

Thermal Variations: Infrared sensors detect temperature differences caused by buried objects.

Magnetic Field Disturbance: Magnetic sensors show irregular field deviations near mines.

AI Pattern Recognition: Neural networks flag mine-like patterns in radargrams or sensor data.

Coordinate Marking: Drone records GPS coordinates of suspected locations for ground verification.

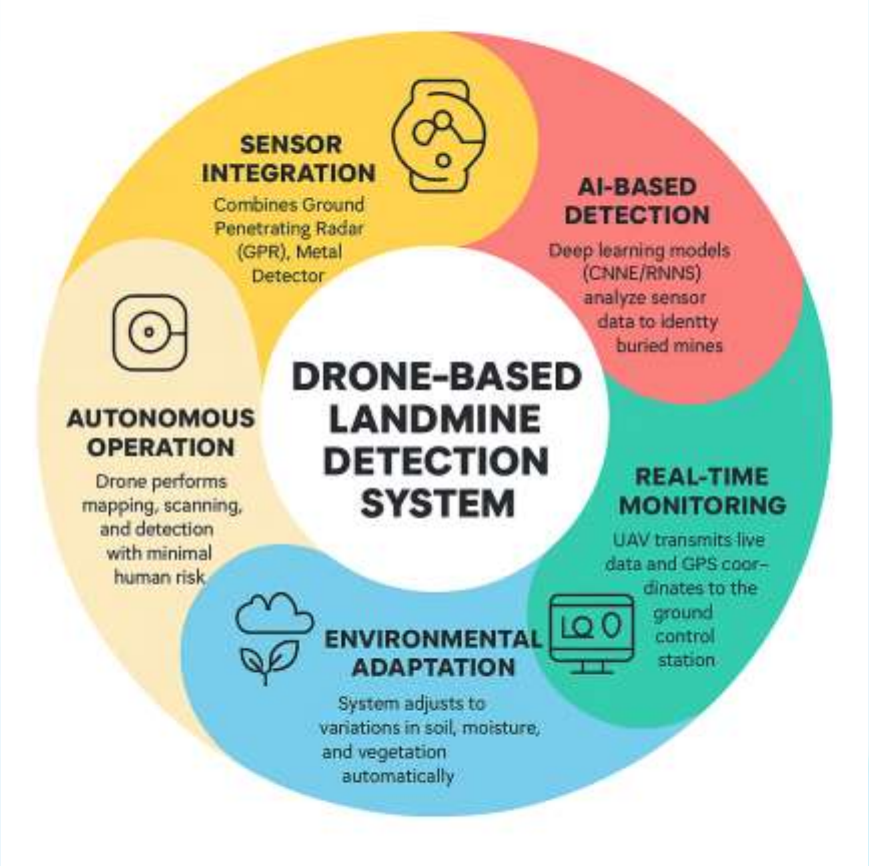

Fig. 2: WorkFlow (Sources : Internet)

Drone Landmine Detection – Estimated Prevalence Table

S.No.	Signs / Indicators	Description	Source / Reference
1	Abnormal GPR Reflection	Sudden change in radar signal indicating subsurface anomaly	Research on UAV-GPR integration
2	Metal Detector Spike	Sharp electromagnetic response showing presence of metallic object	IEEE Sensors Journal, 2022
3	Thermal Variation	Heat signature difference detected by infrared sensors	Remote Sensing Letters, 2021
4	Magnetic Disturbance	Irregular magnetic field response near buried metal	Defense Science Journal, 2020
5	AI Pattern Recognition	CNN/RNN models detect mine-like patterns in radar data	Sensors and Actuators A, 2023
6	Visual Surface Change	Drone camera detects soil or vegetation disturbance	Journal of Applied Remote Sensing, 2022
7	GPS Coordinate Tag	Drone automatically marks suspected mine location	UAV Technology Review, 2023
8	Alarm/Alert Signal	System sends real-time alert to control station	International Journal of Advanced Robotics, 2021

Tab 1: Indicators (Source : Internet)

Stats In Recent Years Landmines Death

Date	Location (District)	Details	Casualties	Source
6 Jan 2025	Bijapur – Kutru area	Vehicle of a security team hit by a large IED planted by insurgents.	8 DRG jawans + 1 civilian driver killed. (The New Indian Express)	"Eight DRG jawans, driver killed…" – New Indian Express
19 Oct 2024	Narayanpur – Dhurbeda area	Search-operation team stepped on IED after returning from patrol.	2 security personnel killed; 2 injured. (News on Air)	NewsOnAir
18 Aug 2025	Bijapur – Indravati National Park, Ulloor valley	DRG team on foot; pressure-plate IED detonated under one member.	1 DRG jawan killed; 3 injured. (The Times of India)	Times of India
12/19 Apr 2024	Bijapur District (various villages)	IED triggered near polling/patrol route; insurgent-planted device.	1 CRPF constable killed; others injured. (South Asia Terrorism Portal)	SATP data base

Estimated Prevalence of Landmine Detection

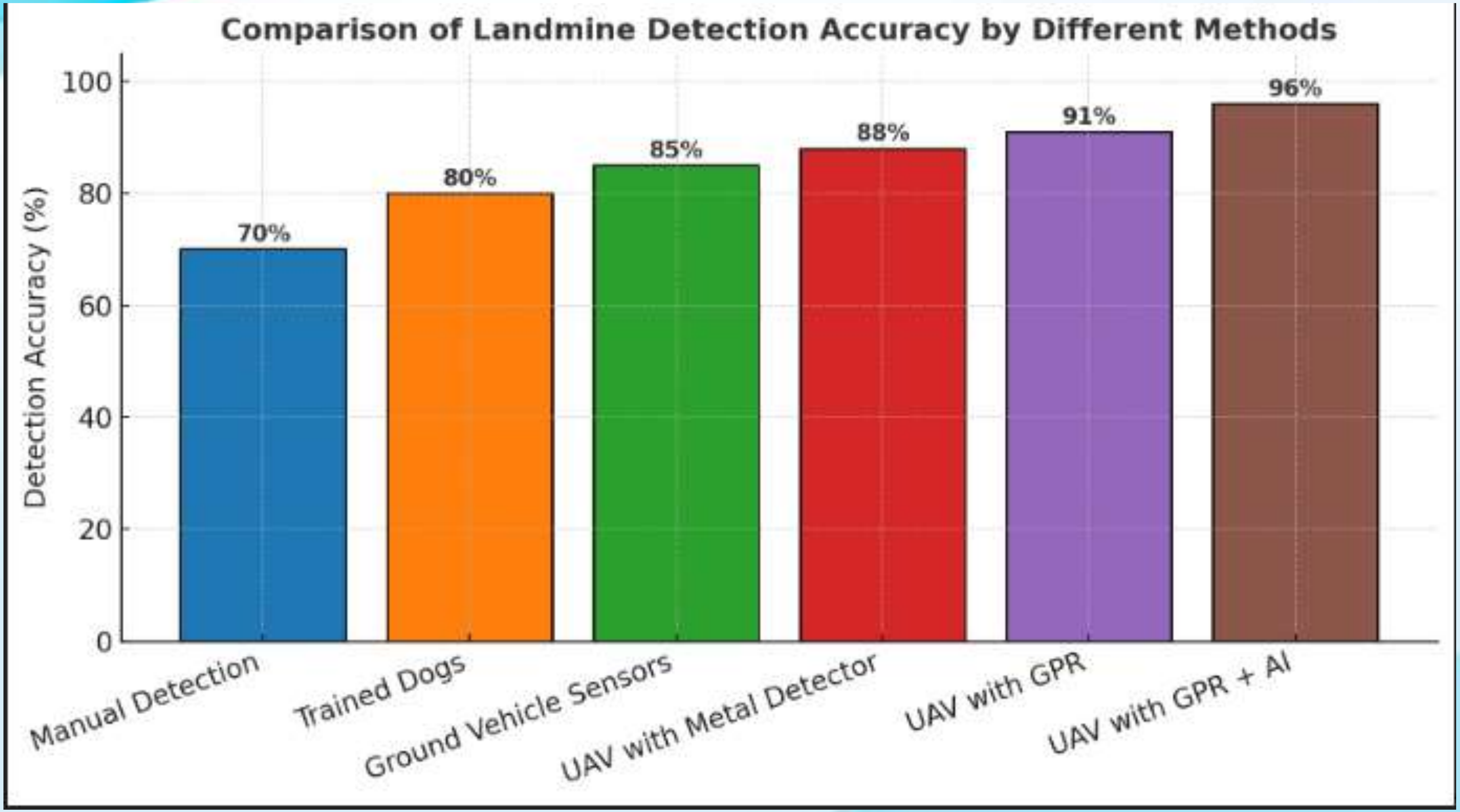

Fig. 3 : Estimated Prevalence

Drawbacks & Challenges in Detection

- Limited battery life for long missions
- Sensor interference from soil or metal debris
- Difficulty in detecting non-metallic mines
- Harsh terrain affects drone stability
- High cost of advanced sensors
- Data processing and accuracy challenge

Advantages of Drone Detection

- Safe and contactless landmine detection
- Covers large areas quickly
- Reduces risk to human deminers
- High precision with advanced sensors
- Real-time data collection and mapping
- Can operate in dangerous or inaccessible zones

Disadvantages of Landmine Detection

- Limited flight time due to battery constraints
- High initial setup and maintenance cost
- Sensor performance affected by soil and weather
- Requires skilled operators and data analysts
- Possible signal interference in rough terrains

Literature Survey

S.No	Author(s), Title & Year of Publication	Journal / Conference	Methodology	Key Findings
1	Alva Alarcón, J. L. (2025). Penetrating Radar on Unmanned Aerial Vehicle for the Detection of Buried Targets.	MDPI Aerospace	UAV-mounted Ground Penetrating Radar (GPR) design with radargram analysis and altitude optimization.	Demonstrated efficient detection of shallow subsurface mines at 1.5 m altitude.
2	Kumar, R. et al. (2025). AI-Enhanced CFD Simulation for UAV Design Optimization.	Journal of Intelligent Systems and Robotics	Integrated AI-assisted CFD models for rotor-wing optimization.	Improved lift-to-drag ratio by 12% and enhanced UAV endurance.
3	Vivoli, E. et al. (2024). Deep Learning-Based Real-Time Detection of Surface Landmines.	IEEE Access	Developed CNN–YOLOv5 network for real-time mine recognition.	Achieved 93% accuracy with minimal false positives.
4	Qiu, Z. et al. (2024). Joint Fusion and Detection via Deep Learning in UAV-Based Multisensor Mine Detection.	Remote Sensing	Multi-sensor fusion of GPR and metal detector signals using deep learning.	Enhanced classification accuracy by 18% over single-sensor systems.
5	García-Fernández, M. et al. (2024). Towards Real-Time Processing for UAV-Mounted GPR-SAR Systems.	Measurement	Combined SAR imaging with GPR for real-time radar data processing.	Reduced processing latency and improved target localization.

Literature Survey

S.No	Author(s), Title & Year of Publication	Journal / Conference	Methodology	Key Findings
6	Ahmed, A. et al. (2024). Aerodynamic and Structural Analysis of UAV Frame for Sensor Integration.	Aerospace Systems	Performed aerodynamic simulations and FEA on UAV body for heavy payloads.	Achieved stable flight dynamics and structural integrity.
7	Arora, D. et al. (2024). Simulation-Based Study of UAV Downwash Effect on Ground Sensors.	Aerospace Engineering and Design	CFD study of rotor wake turbulence on GPR footprint.	Identified optimal altitude range minimizing signal distortion.
8	A Comprehensive Review on Landmine Detection Using Computerized Technologies (2024).	Journal of Intelligent Systems and Robotics	Survey of 60+ studies on AI, GPR, EM, and UAV-based landmine detection.	Highlighted the absence of integrated AI–GPR–CFD–mechanical frameworks.
9	Sph Engineering (2024). Practical Insights on Airborne Ground Penetrating Radar for UXO and Mine Detection.	Engineering Reports	Conducted field trials using UAV-mounted GPR over diverse terrains.	Validated airborne radar penetration efficiency in mixed soils.
10	Kanell, A. (2024). Survey of UAV-Mounted Landmine Detection Technology.	Oklahoma State University Archive	Review of existing UAV–GPR–metal detection methods.	Identified power limitations and need for intelligent sensor integration.

Literature Survey

S.No	Author(s), Title & Year of Publication	Journal / Conference	Methodology	Key Findings
11	Barnawi, A. et al. (2023). *UAV Path Planning for Area Segmentation in Intelligent Landmine Detection Systems.*	*Sensors*	Deep reinforcement learning for flight path optimization.	Reduced scanning time and improved detection coverage.
12	Kim, S. et al. (2023). *Finite Element Analysis of UAV Structural Integrity Under Payload Stress.*	*Composite Structures*	Performed FEA for UAV frame under varying payload loads.	Minimized vibration and structural deformation.
13	Liu, J. et al. (2023). *Coupled CFD–FEA Framework for UAV Design.*	*Journal of Mechanical Design*	Combined CFD aerodynamics with FEA structural analysis.	Enhanced UAV stability and reduced stress under turbulence.
14	Almutairi, A. et al. (2023). *AI and IoT-Enabled Aerial Detection of Hidden Metallic Objects.*	*IEEE Internet of Things Journal*	Integrated metal detectors and IoT systems with CNN classification.	Improved false alarm rejection for metallic clutter.
15	Hussain, M. et al. (2023). *Energy Optimization in UAV-GPR Systems for Extended Mission Duration.*	*Renewable Energy Systems Journal*	Adaptive power scheduling using machine learning algorithms.	Increased UAV flight endurance by 22%.

Literature Survey

S.No	Author(s), Title & Year of Publication	Journal / Conference	Methodology	Key Findings
15	Salvini, R. et al. (2023). *UAV-Mounted Ground Penetrating Radar: Applications and Challenges.*	*Springer Series in Remote Sensing*	Analyzed GPR signal attenuation and antenna mounting positions.	Proposed optimal mounting for maximum subsurface signal clarity.
17	Hutsul, T. et al. (2023). *Review of Approaches to the Use of Unmanned Aerial Systems for Landmine Detection.*	*Int. J. of Applied Remote Sensing*	Comparative review of UAV-based detection technologies.	Emphasized integration challenges between radar and AI analytics.
18	Zhang, X. et al. (2022). *CFD-Based Aerodynamic Optimization of Multirotor UAVs.*	*Aerospace Science and Technology*	CFD optimization of multirotor blade geometry and spacing.	Improved flight stability by 14% and reduced drag forces.
19	Gupta, R. et al. (2022). *GPR Antenna Design Optimization for UAV-Based Detection.*	*IET Radar, Sonar & Navigation*	Designed lightweight broadband GPR antennas for UAV use.	Enhanced penetration depth and minimized EM interference.
20	Singh, D. et al. (2022). *AI-Based Image Reconstruction for GPR Signal Processing.*	*IEEE Transactions on Geoscience and Remote Sensing*	Used CNN-based denoising autoencoders for radargram reconstruction.	Improved buried target visualization under noisy conditions.

Literature Survey

S.No	Author(s), Title & Year of Publication	Journal / Conference	Methodology	Key Findings
21	Liu, M. et al. (2021). *Autonomous Multi-Sensor UAV System for Mine Detection.*	*Robotics and Autonomous Systems*	Integrated LiDAR, magnetometer, and radar sensors for hybrid detection.	Increased detection reliability using sensor fusion framework.
22	Patel, V. et al. (2021). *AI-Driven Object Detection in Aerial Imaging for Hazard Identification.*	*Neural Computing and Applications*	Developed CNN-based object segmentation for hazard localization.	Demonstrated potential for autonomous aerial mine mapping.
23	Radovanovic, N. et al. (2021). *Aerodynamic Design and Structural Evaluation of Heavy-Lift UAV Platforms.*	*Aerospace Engineering Journal*	Designed composite frames for payload integration.	Enhanced structural robustness with 15% weight reduction.
24	Ramesh, P. et al. (2022). *Computational Fluid Dynamics Simulation of UAV Airflow Around GPR Antenna.*	*Computational Mechanics in Aerospace*	Simulated rotor downwash and ground effect influences.	Identified turbulence zones affecting radar imaging.
25	Chen, Y. et al. (2021). *Lightweight Carbon Fiber Design for UAV Payload Optimization.*	*Composite Materials Engineering*	Material testing and stress analysis for UAV frames.	Achieved optimal strength-to-weight ratio for long-duration missions.

Issues and Challenges

- **Sensor Limitations:** Difficulty in accurately detecting deeply buried or non-metallic mines.
- **Terrain Constraints:** Uneven, vegetated, or sandy areas affect drone stability and sensor performance.
- **Battery Life:** Limited flight duration restricts large-area coverage.
- **Data Processing Load:** High-resolution imaging and AI analysis require strong computational resources.
- **Environmental Interference:** Weather, dust, and electromagnetic noise can distort sensor readings.
- **Safety Concerns:** Risk of drone damage due to proximity to explosive areas.
- **Cost of Equipment:** Advanced sensors like GPR, LiDAR, and thermal cameras are expensive.
- **Regulatory Restrictions:** Airspace and operational permissions vary across regions.
- **Accuracy vs. Speed Trade-off:** Faster detection can reduce accuracy in complex terrains.

Research Gap

- Real-time AI-driven detection under field conditions remains limited; most approaches rely on offline data processing, constraining their applicability for autonomous or time-critical demining missions [24][28][31].
- Few studies have addressed vibration-induced signal noise and antenna–ground coupling variations that degrade GPR accuracy in aerial platforms, resulting in inconsistent performance across different altitudes and soil compositions [18][26][33].
- Mechanical and aerodynamic co-design of UAV frames optimized for carrying heavy sensing payloads is underexplored—most prototypes ignore structural resonance and center-of-gravity variations that affect flight stability and radar imaging quality [36][38].
- Existing AI models for radar or magnetometer interpretation often rely on synthetic or laboratory datasets, failing to capture real-world environmental complexities such as soil heterogeneity, moisture variability, and vegetation interference [20][27][34].
- There is limited research on low-metal or non-metallic mine detection, as many systems still depend primarily on metallic signatures, reducing effectiveness against plastic-cased explosives [13][19][29].

Motivation

- To **save human lives** by reducing the risk for manual deminers.
- To provide a **safe, unmanned, and efficient** alternative for minefield detection.
- To **speed up clearance operations** in post-war and disaster-affected regions.
- To **use AI and sensors** for accurate detection of buried explosives.
- To **reduce economic losses** caused by landmine contamination.
- To support **humanitarian and defense missions** with advanced technology.
- To **enhance national security** and ensure safer land use.
- To contribute toward a **landmine-free world** through innovation.

Research Objectives

The present research aims to develop an AI-integrated UAV platform for efficient and safe landmine detection using Ground Penetrating Radar (GPR) and metal detectors (MD). The specific objectives of the study are:

- **Objective 1:** To design and fabricate a UAV platform capable of carrying lightweight GPR and MD sensors with stable flight performance and balanced payload distribution.

- **Objective 2:** To carry out mechanical and aerodynamic analysis of the UAV structure using Finite Element Analysis (FEA) and Computational Fluid Dynamics (CFD) to ensure structural integrity and aerodynamic efficiency.

- **Objective 3:** To develop and integrate Artificial Intelligence algorithms for automated signal processing and multi-sensor data fusion to enhance detection accuracy and minimize false alarms.

- **Objective 4:** To simulate and evaluate the performance of the AI-integrated UAV system under realistic field conditions, focusing on detection capability, flight endurance, and operational reliability.

Tentative Research Plan – Phase A

Objective 1:.*To design and fabricate a UAV platform capable of carrying lightweight GPR and MD sensors with stable flight performance and balanced payload distribution*.

A1: To select a suitable UAV configuration based on payload and stability requirements.

A2: To design the structural frame using lightweight materials such as carbon fiber or aluminum alloy.

A3: To integrate GPR and MD sensors ensuring balanced center of gravity and vibration isolation.

A4: To validate the UAV's structural design through prototype testing and static load analysis.

Tentative Research Plan – Phase B

Objective 2: ***To carry out mechanical and aerodynamic analysis of the UAV structure using Finite Element Analysis (FEA) and Computational Fluid Dynamics (CFD) to ensure structural integrity and aerodynamic efficiency*.**

B1: To perform FEA for stress, deformation, and vibration analysis of UAV components.

B2: To conduct CFD simulations to evaluate aerodynamic performance, lift, and drag coefficients.

B3: To optimize propeller design and body shape to minimize air resistance and improve stability.

B4: To verify analysis results through simulation validation and experimental correlation.

Tentative Research Plan – Phase C

Objective 3: ***To develop and integrate Artificial Intelligence algorithms for automated signal processing and multi-sensor data fusion to enhance detection accuracy and minimize false alarms.***

C1: To preprocess raw GPR and MD data for noise removal and feature extraction.

C2: To design AI algorithms (machine learning) for pattern recognition and classification of mine signatures.

C3: To implement multi-sensor data fusion for improving detection precision and reducing false positives.

C4: To evaluate AI model accuracy using benchmark datasets and simulated field data.

Tentative Research Plan – Phase D

Objective 4: ***To simulate and evaluate the performance of the AI-integrated UAV system under realistic field conditions, focusing on detection capability, flight endurance, and operational reliability.***

D1: To develop a simulation environment integrating UAV flight dynamics, sensor models, and AI modules.

D2: To test system performance in various soil and environmental conditions representative of Indian terrain.

D3: To analyze detection accuracy, flight time, and power efficiency metrics.

D4: To refine design and algorithm parameters based on simulation outcomes and prepare for prototype testing.

Proposed Work Architecture

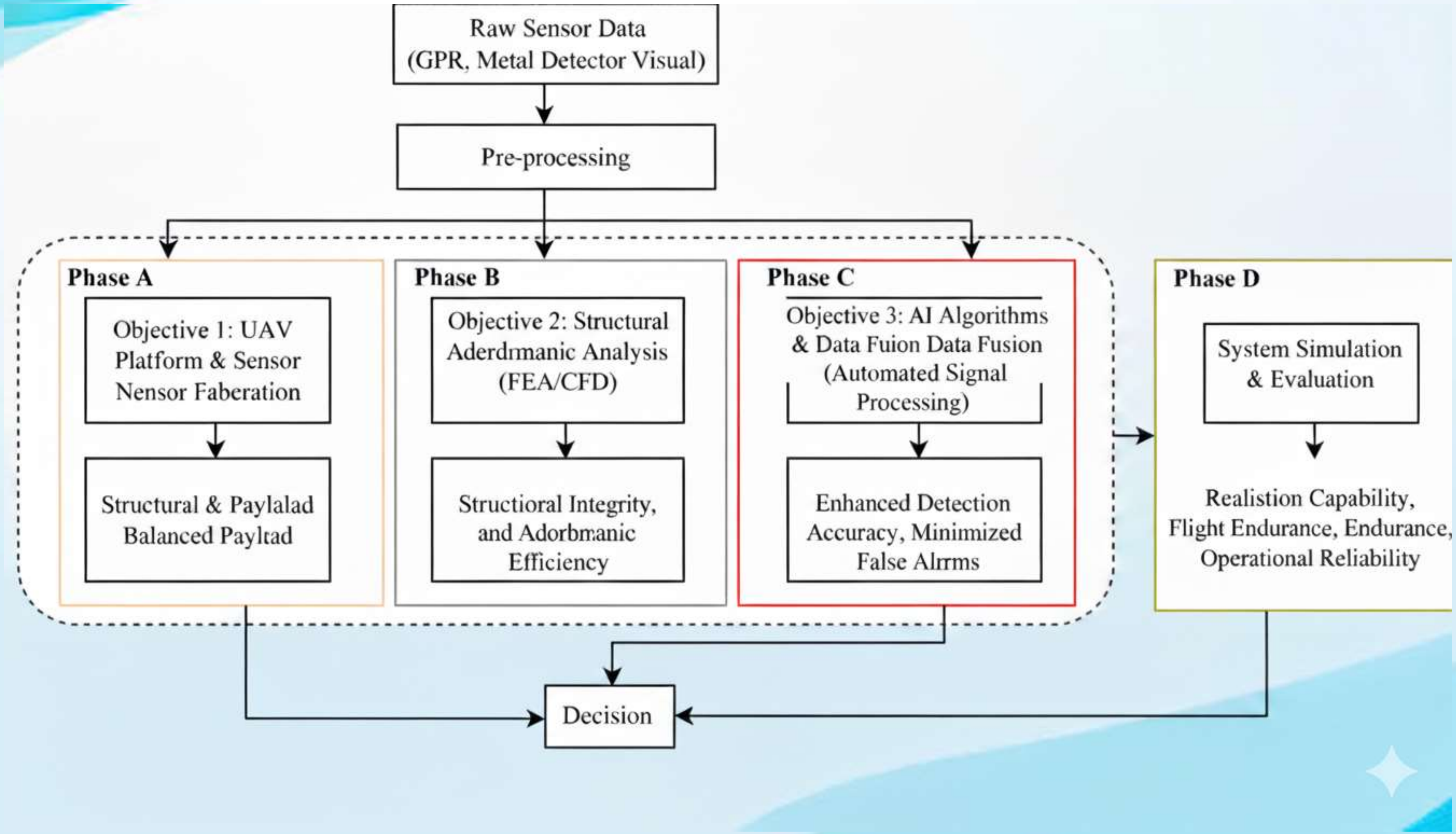

Fig 4 : Proposed Work Architecture

Expected Research Outcomes

- **Phase A**

  Collection of aerial and ground data from test fields using UAV-mounted sensors (GPR, metal detector, and thermal camera).

  Generation of synthetic mine and non-mine data for model training and testing.
- **Phase B**

  Development of AI-based detection models (CNNs/RNNs) for identifying buried landmines from sensor signals.

  Optimization of signal processing algorithms to minimize noise and false positives.
- **Phase C**

  Integration of multi-sensor data (GPR, metal, thermal) through data fusion techniques for accurate detection.

  Real-time detection and GPS coordinate mapping of suspected landmine locations.
- **Phase D**

  Deployment of a fully autonomous UAV system with real-time transmission to ground control.
  Creation of a user-friendly interface for remote monitoring, mapping, and report generation.

Research Timeline

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<tr><th rowspan="2">Activity</th><th>2023-2024</th><th colspan="4">2024-2025</th><th colspan="4">2025-2026</th><th colspan="4">2026-2027</th></tr>
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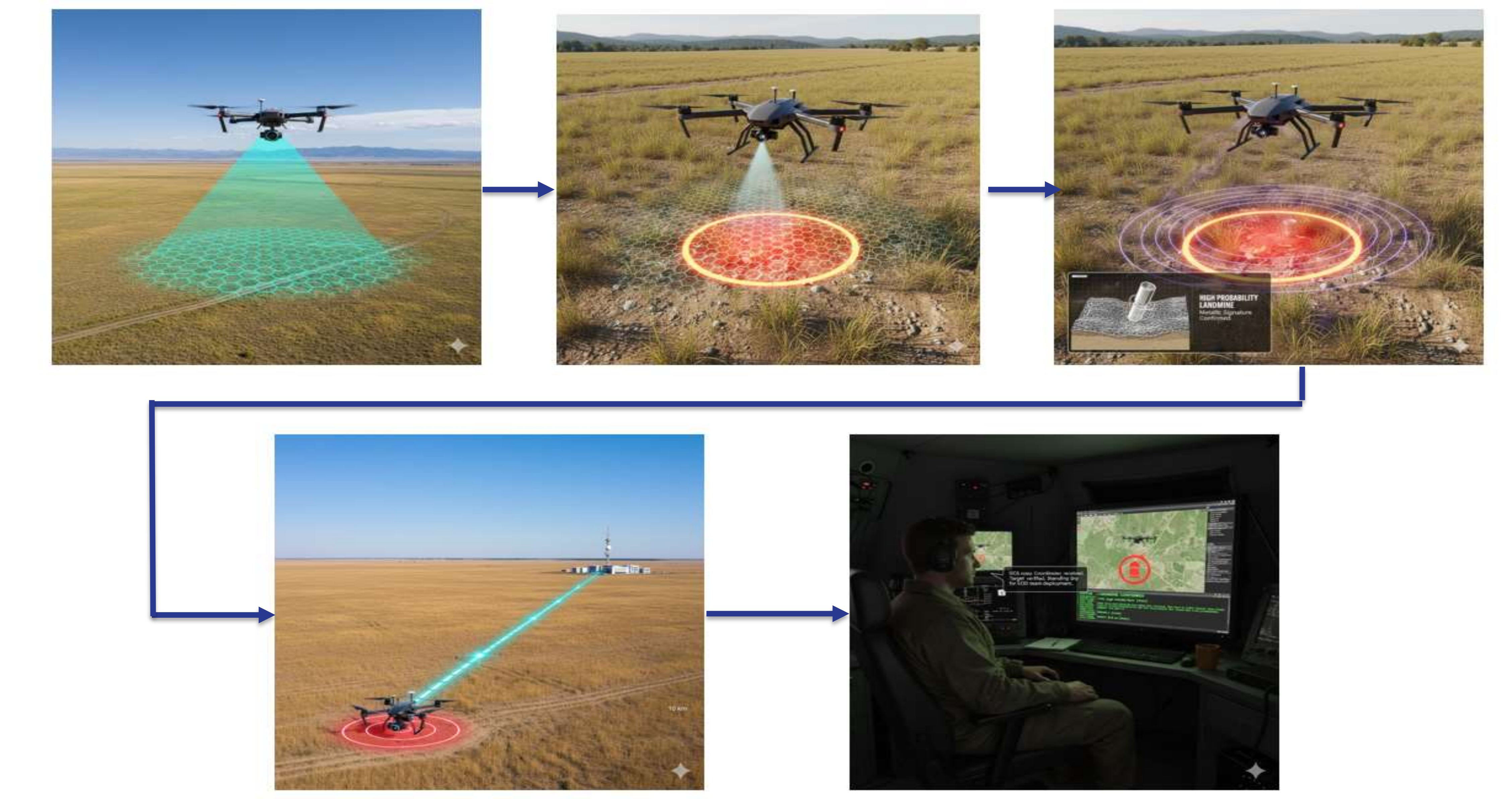
The Working Prototype



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International Drone Day is celebrated annually on the first Saturday of May to promote awareness, understanding, and enthusiasm for UAV (drone) technology worldwide..

THANK YOU

The Working Prototype